

LECTURE 2: DC MOTOR DRIVES

DC motor drives have been widely used in applications requiring adjustable speed; good speed regulation; and frequent starting, braking, and reversing. Various DC motor drives have been widely applied to different electric traction applications because of their technological maturity and control simplicity.

Principle of Operation and Performance

The operation principle of a DC motor is straight forward. When a wire carrying electric current is placed into a magnetic field, a magnetic force acting on the wire is produced. The force is perpendicular to the wire and the magnetic field as shown in Figure 1.

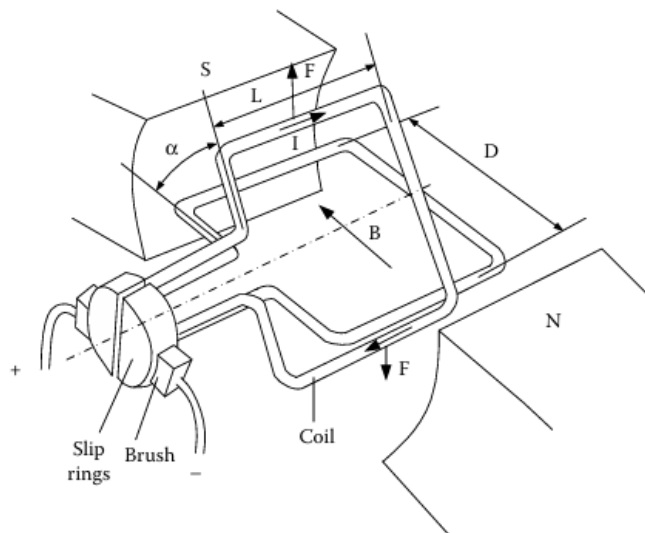


Figure 1: Operation principle of a DC motor.

The magnetic force is proportional to the wire length, magnitude of the electric current, and the density of the magnetic field; that is,

$$F = BIL$$

When the wire is shaped into a coil, the magnetic forces acting on both sides produce a torque, which is expressed as

$$T = BIL \cos \alpha$$

Where α is the angle between the coil plane and the magnetic field as shown in Figure 1. The magnetic field may be produced by a set of windings or PMs. The former is called wound-field DC motor and the latter is called the PM DC motor. The coil carrying electric current is called the armature. In practice, the armature consists of a number of coils. In order to obtain continuous and maximum torque, slip rings and brushes are used to conduct each coil at the position of $\alpha = 0$

Practically, the performance of DC motors can be described by the armature voltage, back electromotive force (EMF), and field flux. Typically, there are four types of wound-field DC motors, depending on the mutual interconnection between the field and armature windings. They are separately excited, shunt excited, series excited, and compound excited as shown in Figure.

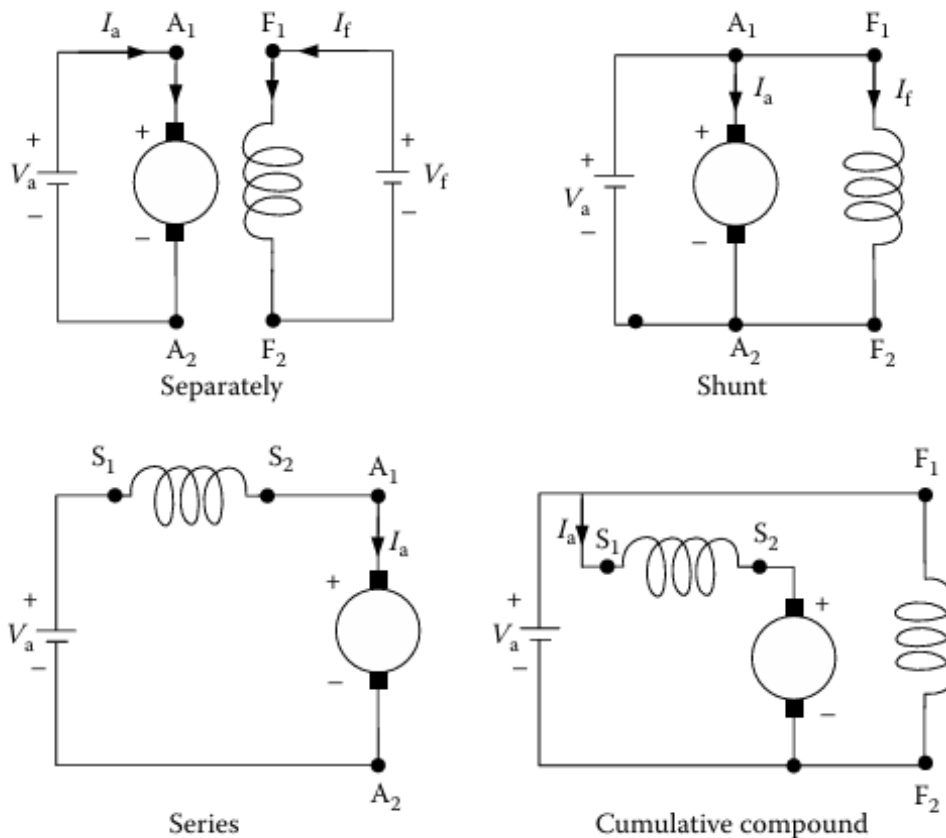


Figure 2: Wound-field DC motors.

In the case of a separately excited motor, the field and armature voltage can be controlled independently of one another. In a shunt motor, the field and armature are connected in parallel to a common source. Therefore, an independent control of field or armature currents can only be achieved by inserting a resistance in the appropriate circuit. This is an inefficient method of control. The efficient method is to use power electronics based DC-DC converters in the appropriate circuit to replace the resistance. The DC-DC converters can be actively controlled to produce proper armature and field voltage. In the case of a series motor, the field current is the same as the armature current; therefore, field flux is a function of armature current. In a cumulative compound motor, the magnetomotive force (mmf) of a series field is a function of the armature current and is in the same direction as the mmf of the shunt field.

The steady-state equivalent circuit of the armature of a DC motor is shown in Figure 3.

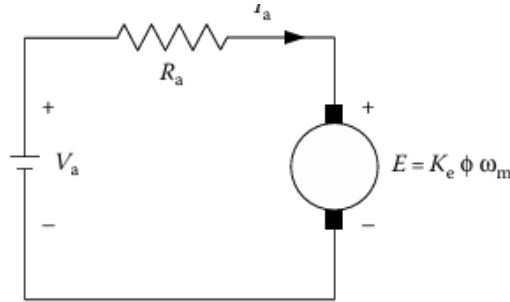


Figure 3: Steady-state equivalent circuit of the armature circuit of a DC motor.

The resistor R_a is the resistance of the armature circuit. For separately excited and shunt DC motors, it is equal to the resistance of the armature windings; for the series and compound motors, it is the sum of armature and series field winding resistances. The basic equations of a DC motor are

$$V_a = E + R_a I_a \quad \text{and} \quad E = K_e \phi \omega_m$$

$$T = K_e \phi I_a$$

where ϕ is the flux per pole in webers, I_a is the armature current in A, V_a is the armature voltage in volts, R_a is the resistance of the armature circuit in ohms, ω_m is the speed of the armature in rad/s, T is the torque developed by the motor in N m, and K_e is a constant

From Equations, one can obtain

$$T = \frac{K_e \phi}{R_a} V - \frac{(K_e \phi)^2}{R_a} \omega_m$$

Equations are applicable to all the DC motors, namely separately (or shunt) excited, series, and compound motors. In the case of separately excited motors, if the field voltage is maintained as constant, one can assume the flux to be practically constant as the torque changes. In this case, the speed–torque characteristic of a separately excited motor is a straight line, as shown in Figure 4.

The non load speed ω_m is determined by the armature voltage and the field excitation. Speed decreases as torque increases, and speed regulation depends on the armature circuit resistance. Separately excited motors are used in applications requiring good speed regulation and proper adjustable speed.

In the case of series motors, the flux is a function of armature current. In an unsaturated region of the magnetization characteristic, ϕ can be assumed to be proportional to I_a . Thus

$$\phi = K_f I_a$$

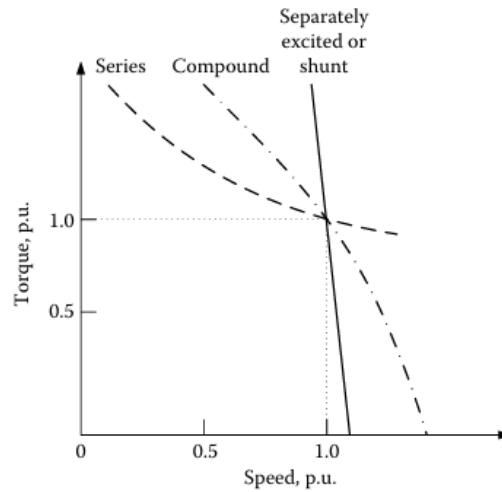


Figure 4: Speed characteristics of DC motors

By Equations $T = K_e \phi I_a$ through above equation, the torque for series excited DC motors can be obtained as

$$T = \frac{K_e K_f V_a^2}{(R_a + K_e K_f \omega_m)^2}$$

where the armature circuit resistance R_a is now the sum of armature and field winding resistance.

A speed–torque characteristic of a series DC motor is shown in Figure. In the case of a series, any increase in torque is accompanied by an increase in the armature current and, therefore, an increase in magnetic flux. Because flux increases with the torque, the speed drops to maintain a balance between the induced voltage and the supply voltage. The characteristic, therefore, shows a dramatic drop. A motor of standard design works at the knee point of the magnetization curve at the rated torque. At heavy torque (large current) over load, the magnetic circuit saturates and the speed–torque curve approaches a straight line.

Series DC motors are suitable for applications requiring high starting torque and heavy torque overload, such as traction. This was just the case for electric traction before the power electronics and micro-control era. However, series DC motors for traction application bear some disadvantages. They are not allowed to operate without the load torque with full supply voltage. Otherwise, their speed will quickly increase up to a very high value. Another disadvantage is the difficulty of regenerative braking. Performance equations for cumulative compound DC motors can be derived from Equations 3 and 4. The speed–torque characteristics are between series and separately excited (shunt) motors, as shown in Figure 4.

Combined Armature Voltage and Field Control

The independence of armature voltage and field provides more flexible control of the speed and torque than other types of DC motors. In EV and HEV applications, the most desirable speed–torque characteristic is to have a constant torque below a certain speed (base speed), and a constant power in the speed range of above the base speed, as shown in Figure 5.

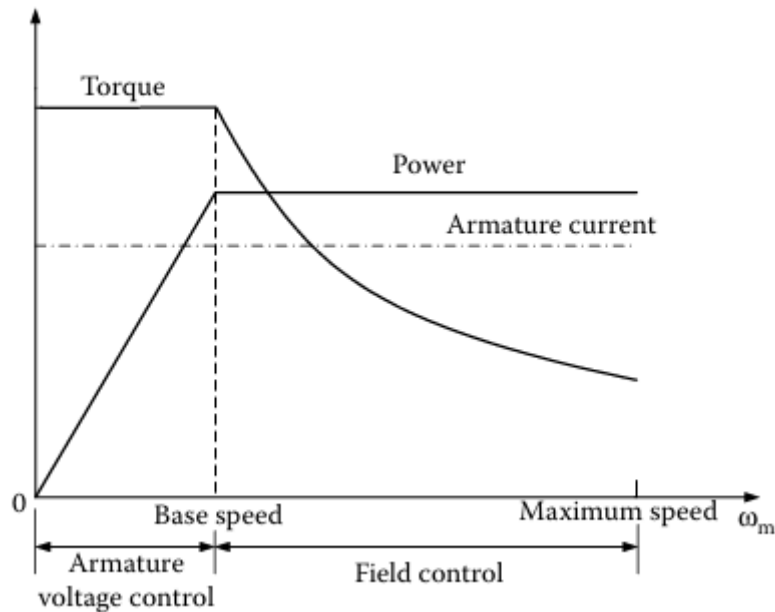


Figure 5: Torque and power limitations in combined armature voltage and field control

In the speed range of lower than the base speed, the armature current and field are set at their rated values, producing the rated torque. From Equations 3 through 4, it is clear that the armature voltage must be increased proportionally with the increase of the speed. At the base speed, the armature voltage reaches its rated value (equal to the source voltage) and cannot be increased further. In order to further increase the speed, the field must be weakened with the increase of the speed, then maintaining the back EMF E and armature current constant. The torque produced drops parabolically with the increase of the speed and the output power remains constant, as shown in Figure 5.

Chopper Control of DC Motors

Choppers are used for the control of DC motors because of a number of advantages such as high efficiency, flexibility in control, light weight, small size, quick response, and regeneration down to very low speeds. At present, the separately excited DC motors are usually used in traction, due to the control flexibility of armature voltage and field.

For a DC motor control in open-loop and closed-loop configurations, the chopper offers a number of advantages due to the high operation frequency. High operation frequency results in high-frequency output voltage ripple and therefore less ripples in the motor armature current and a smaller region of discontinuous conduction in the speed–torque plane. A reduction in the armature current ripple reduces the armature losses. A reduction or elimination of the discontinuous conduction region improves speed regulation and transient response of the drive.

The power electronic circuit and the steady-state waveform of a DC chopper drive are shown in Figure 6.

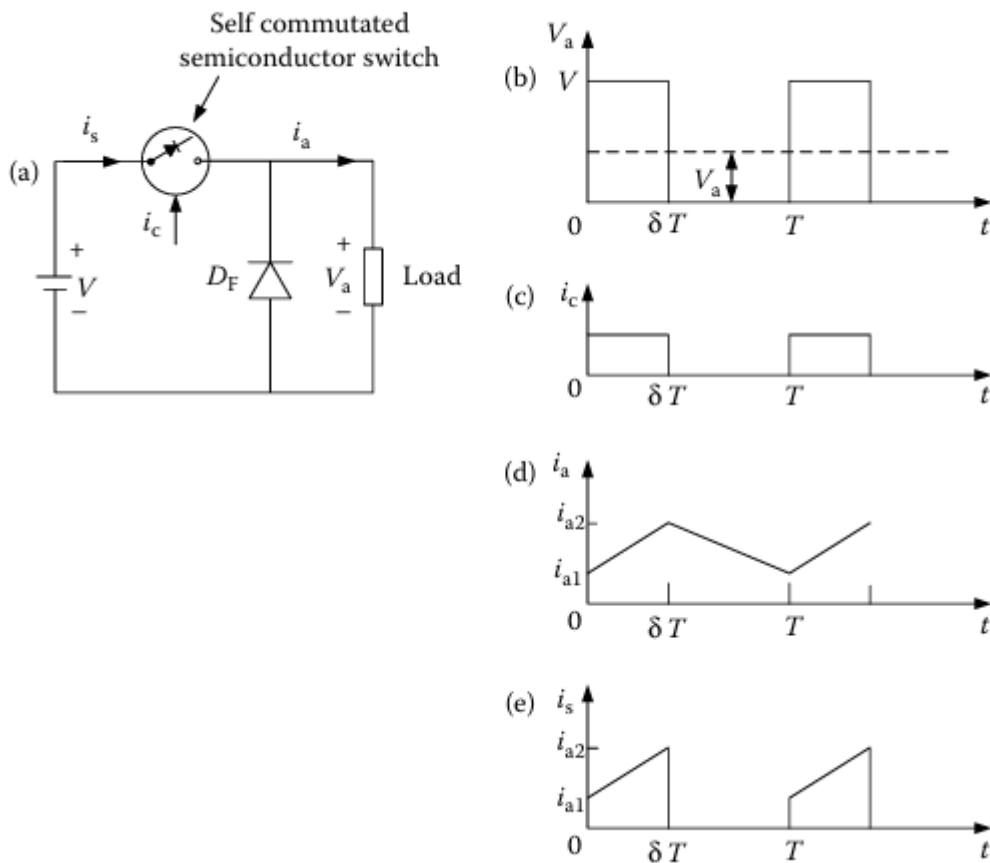


Figure 6: Principle of operation of a step down (or class A) chopper: (a) basic chopper circuit; (b)–(e) waveforms

A DC voltage source, V , supplies an inductive load through a self-commutated semiconductor switch S . The symbol of a self commutated semiconductor switch has been used because a chopper can be built using any devices among thyristors with a forced commutation circuit: GTO, power transistor, MOSFET, and IGBT. The diode shows the direction in which the device can carry current. A diode D_F is connected in parallel with the load. The semiconductor switch S is operated periodically over a period T and remains closed for a time $t_{on} = \delta T$ with $0 < \delta < 1$. The variable $\delta = \frac{t_{on}}{T}$ is called the duty ratio or duty cycle of a chopper. Figure also shows the waveform of control signal i_c . Control signal i_c will be the base current for a transistor chopper, and a gate current for the GTO of a GTO chopper or the main thyristor of a thyristor chopper. If a power MOSFET is used, it will be a gate to the source voltage. When the control signal is present, the semiconductor switch S will conduct, if forward biased. It is assumed that the circuit operation has been arranged such that the removal of i_c will turn off the switch.

During the on interval of the switch ($0 \leq t \leq \delta T$), the load is subjected to a voltage V and the load current increases from i_{a1} to i_{a2} . The switch is opened at $t = \delta T$. During the off period of the switch ($\delta T \leq t \leq T$), the load inductance maintains the flow of current through diode D_F . The load terminal voltage stays zero (if the voltage drop on the diode is ignored in comparison to V) and the current decreases from i_{a2} to i_{a1} . The interval $0 \leq t \leq \delta T$ is called the duty interval and the interval $\delta T \leq t \leq T$ is known as the freewheeling interval. Diode D_F provides a path for the load current to flow when switch S is off, and thus improves the load current waveform. Furthermore, by maintaining the continuity of the load current at turn-off, it prevents transient voltage from appearing across switch S , due to the sudden change of the load current. The source current waveform is also shown in Figure 6e. The source current flows only during the duty interval and is equal to the load current. The direct component or average value of the load voltage V_a is given by

$$V_a = \frac{1}{T} \int_0^T v_a dt = \frac{1}{T} \int_0^{\delta T} V dt = \delta V$$

By controlling δ between 0 and 1, the load voltage can be varied from 0 to V ; thus a chopper allows a variable DC voltage to be obtained from a fixed voltage DC source.

The following important points can be noted from the waveform of Figure 6.

1. The source current is not continuous but flows in pulses. The pulsed current makes the peak input power demand high and may cause fluctuation in the source voltage. The source current waveform can be resolved into DC and AC harmonics. The fundamental AC harmonic frequency is the same as the chopper frequency. The AC harmonics are undesirable because they interfere with other loads connected to the DC source and cause radio frequency interference through conduction and electromagnetic radiation. Therefore, an L-C filter is usually incorporated between the chopper and the DC source. At higher chop per frequencies, harmonics can be reduced to a tolerable level

by a cheaper filter. From this point, a chopper should be operated at the highest possible frequency.

2. The load terminal voltage is not a perfect direct voltage. In addition to a direct component, it has the harmonics of the chopping frequency and its multiples. The load current also has an AC ripple.

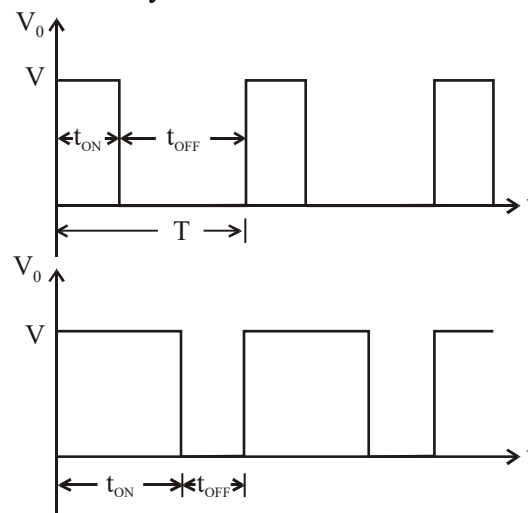
The chopper of Figure 6 is called a class A chopper. It is one of the number of chopper circuits that are used for the control of DC motors. This chopper is capable of providing only a positive voltage and a positive current. It is called a single-quadrant chopper, only providing separately excited DC motor control in the first quadrant, that is, positive speed and positive torque. Since it can vary the output voltage from V to 0 , it is also a step-down chopper or a DC to-DC buck converter. The basic principle involved can also be used to realize a step-up chopper or DC-to-DC boost converter.

The switch S can be controlled in various ways for varying the duty ratio δ . The control technologies can be divided into two categories:

1. Time Ratio Control (TRC).
2. Current Limit Control (CLC).

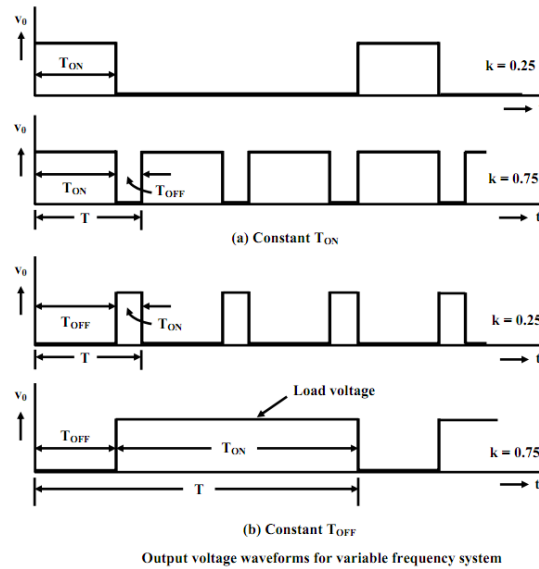
In TRC, also known as pulse width control, the ratio of on time to chopper period is controlled. The TRC can be further divided as follows:

1. Constant frequency TRC: The chopper period T is kept fixed and the on period of the switch is varied to control the duty ratio δ .



2. Varied frequency TRC: Here δ is varied either by keeping t_{on} constant and varying T or by varying both t_{on} and T .

In variable frequency control with constant on time, low-output voltage is obtained at very low chopper frequencies. The operation of a chopper at low frequencies adversely affects the motor performance. Furthermore, the operation of a chopper with variable frequencies makes the design of an input filter very difficult. Thus, variable frequency control is rarely used.

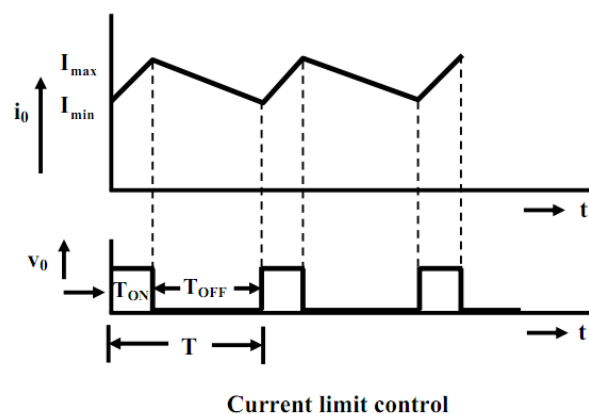


Disadvantages of Variable Frequency Control Strategy

- The frequency has to be varied over a wide range for the control of output voltage in frequency modulation. Filter design for such wide frequency variation is, therefore, quite difficult.
- For the control of a duty ratio, frequency variation would be wide. As such, there is a possibility of interference with systems using certain frequencies, such as signaling and telephone line, in frequency modulation technique.
- The large OFF time in frequency modulation technique, may make the load current discontinuous, which is undesirable.

Thus, the constant frequency system using PWM is the preferred scheme for dc-dc converters (choppers).

In CLC, also known as point-by-point control, δ is controlled indirectly by controlling the load current between certain specified maximum and minimum values. When the load current reaches a specified maximum value, the switch disconnects the load from the source and reconnects it when the current reaches a specified minimum value. For a DC motor load, this type of control is, in effect, a variable frequency variable on time control.



The circuit diagram and steady-state waveforms of a step-up chopper are shown in Figure 7. This chopper is known as a class B chopper. The presence of control signal i_c indicates the duration for which the switch can conduct if forward-biased. During a chopping period T , it remains closed for an interval $0 \leq t \leq \delta T$ and remains open for an interval $\delta T \leq t \leq T$. During the on period, i_s increases from i_{s1} to i_{s2} , thus increasing the magnitude of energy stored in inductance L . When the switch is opened, current flows through the parallel combination of the load and capacitor C . Since the current is forced against the higher voltage, the rate of change of the current is negative. It decreases from i_{s2} to i_{s1} in the switch's off period. The energy stored in the inductance L and the energy supplied by the low-voltage source are given to the load. The capacitor C serves two purposes. At the instant of opening of switch S , the source current, i_s , and load current, i_a , are not the same. In the absence of C , the turn-off of S will force the two currents to have the same values. This will cause high induced voltage in the inductance L and the load inductance. Another reason for using capacitor C is to reduce the load voltage ripple. The purpose of the diode D is to prevent any flow of current from the load into switch S or source V .

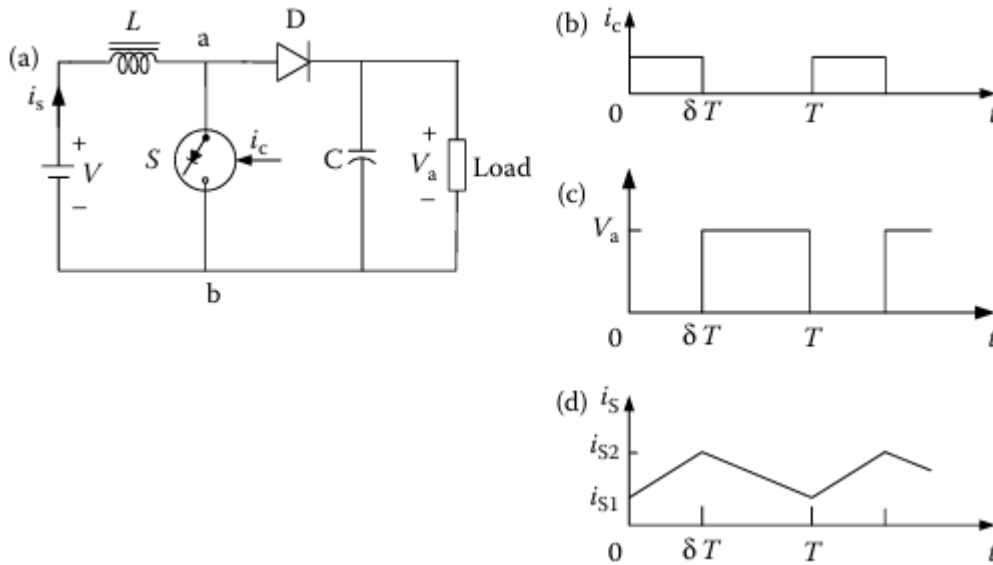


Figure 7: Principle of operation of a step-up (or class B) chopper: (a) basic chopper circuit; (b)-(d) waveforms

For understanding of the step-up operation, capacitor C is assumed large enough to maintain a constant voltage V_a across the load. The average voltage across the terminals a and b is given as

$$V_{ab} = \frac{1}{T} \int_0^T v_{ab} dt = V_a(1 - \delta)$$

The average voltage across the inductance L is

$$V_L = \frac{1}{T} \int_0^T \left(L \frac{di}{dt} \right) dt = \frac{1}{T} \int_{i_{s1}}^{i_{s2}} L di = 0$$

The source voltage

$$V = V_L + V_{ab}$$

Substituting from equations for V_L and V_{ab} into above equation gives

$$V = V_a(1 - \delta) \text{ or } V_a = \frac{V}{1 - \delta}.$$

According to above equations, theoretically the output voltage V_a can be changed from V to ∞ by controlling δ from 0 to 1. In practice, V_a can be controlled from V to a higher voltage, which depends on the capacitor C , and the parameters of the load and chopper. The main advantage of a step-up chopper is the low ripple in the source current. While most applications require a step-down chopper, the step-up chopper finds application in low-power battery-driven vehicles. The principle of the step-up chopper is also used in the regenerative braking of DC motor drives

Multi-Quadrant Control of Chopper-Fed DC Motor Drives

The application of DC motors on EVs and HEVs requires the motors to operate in multi-quadrants, including forward motoring, forward braking, backward motoring, and backward braking, as shown in Figure 8. For vehicles with reverse mechanical gears, two-quadrant operation (forward motoring and forward braking, or quadrant I and quadrant IV) is required. However, for the vehicles without reverse mechanical gears, four-quadrant operation is needed. Multi-quadrant operation of a separately excited DC motor is implemented by controlling the voltage poles and magnitude through power electronics-based choppers.

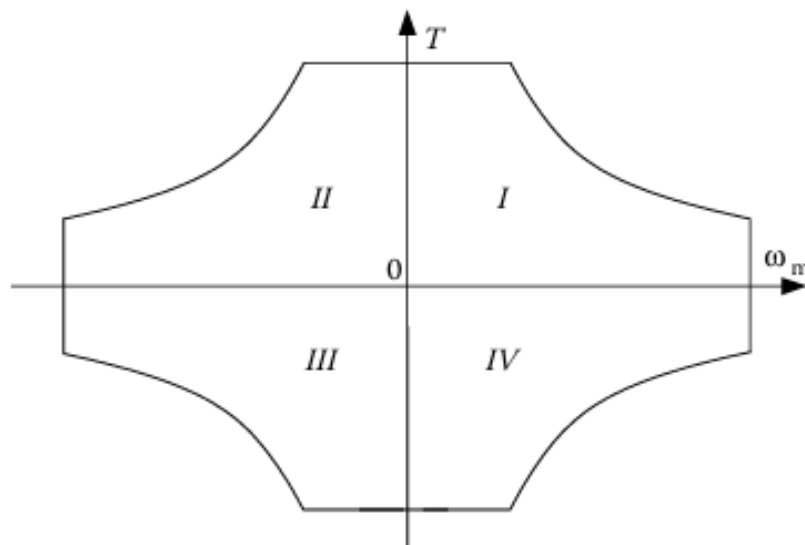


Figure 8: Speed–torque profiles of a multi-quadrant operation

Two-Quadrant Control of Forward Motoring and Regenerative Braking

A two-quadrant operation consisting of forward motoring and forward regenerative braking requires a chopper capable of giving a positive voltage and current in either direction. This two-quadrant operation can be realized in the following two schemes.

Single Chopper with a Reverse Switch

The chopper circuit used for forward motoring and forward regenerative braking is shown in Figure 9 where S is a self-commutated semiconductor switch, operated periodically such that it remains closed for a duration of δT and remains open for the duration of $(1 - \delta)T$. C is the manual switch. When C is closed and S is in operation, the circuit is similar to that of Figure 4, permitting the forward motoring operation. Under this condition, terminal a is positive and terminal b is negative.

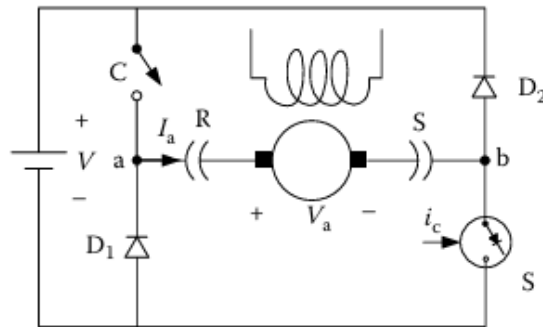


Figure 9: Forward motoring and regenerative braking control with a single chopper

The regenerative braking in the forward direction is obtained when C is opened and the armature connection is reversed with the help of the reversing switch RS , making terminal b positive and terminal a negative. During the on period of the switch S , the motor current flows through a path consisting of the motor armature, switch S , and diode D_1 , and increases the energy stored in the armature circuit inductance. When S is opened, the current flows through the armature diode D_2 , source V , diode D_1 and back to the armature, thus feeding energy into the source. During motoring, the change over to regeneration is done in the following steps. Switch S is deactivated and switch C is opened. This forces the armature current to flow through diode D_2 , source V and diode D_1 . The energy stored in the armature circuit is fed back to the source and the armature current falls to zero. After an adequate delay to ensure that the current has indeed become zero, the armature connection is reversed and switch S is reactivated with a suitable value of δ to start regeneration.

Class C Two-Quadrant Chopper

In some applications, a smooth transition from motoring to braking and vice versa is required. For such applications, the class C chopper is used as shown in Figure 6.10. The self-commutated semiconductor switch S_1 and diode D_1 constitute one chopper and the self-commutator switch S_2 , and diode D_2 form another chopper. Both the choppers are controlled simultaneously, both for motoring and regenerative braking. The switches S_1 and S_2 are closed alternately. In the chopping period T , S_1 is kept on for a duration δT and S_2 is kept on from δT to T . To avoid a direct, short-circuit across the source, care is taken to ensure that S_1 and S_2 do not conduct at the same time. This is generally achieved by providing some delay between the turn-off of one switch and the turn-on of another switch.

The waveforms of the control signals, v_a , i_a , and i_s and the devices under conducting during different intervals of a chopping period are shown in Figure 6.9b. In drawing these waveforms, the delay between the turn-off of one switch and turn-on of another switch has been ignored because it is usually very small. The control signals for the switches S_1 and S_2 are denoted by ic_1 and ic_2 , respectively. It is assumed that a switch conducts only when the control signal is present and the switch is forward biased.

The following points are helpful in understanding the operation of this two-quadrant circuit.

1. In this circuit, discontinuous conduction does not occur, irrespective of its frequency of operation. The discontinuous conduction occurs when the armature current falls to zero and remains zero for a finite interval of time. The current may become zero either during the freewheeling interval or in the energy transfer interval. In this circuit, freewheeling will occur when S_1 is off and the current is flowing through D_1 . This will happen in interval $\delta T \leq t \leq T$, which is also the interval for which S_2 receives the control signal. If i_a falls to zero in the freewheeling interval, the back EMF will immediately drive a current through S_2 in the reverse direction, thus preventing the armature current from remaining zero for a finite interval of time. Similarly, the energy transfer will be present when S_2 is off and D_2 is conducting—that is, during the interval $0 \leq t \leq \delta T$. If the current falls to zero during this interval, S_1 will conduct immediately because ic_1 is present and $V > E$. The armature current will flow, preventing discontinuous conduction.
2. Since discontinuous conditions are absent, the motor current will be flowing all the time. Thus, during the interval $0 \leq t \leq \delta T$, the motor armature will be connected either through S_1 or D_2 . Consequently, the motor terminal voltage will be V and the rate of change of i_a will be positive because $V > E$. Similarly, during the interval $\delta T \leq t \leq T$, the motor armature will be shorted either through D_1 or S_2 . Consequently, the motor voltage will be zero and the rate of change of i_a will be negative.
3. During the interval $0 \leq t \leq \delta T$, the positive armature current is carried by S_1 and the negative armature current is carried by D_2 . The source current flows only during this interval and it is equal to i_a . During the interval $\delta T \leq t \leq T$, the positive current is carried by D_1 and the negative current is carried by S_2 .

4. From the motor terminal voltage waveform of Figure 10b, $V_a = \delta V$.
Hence,

$$I_a = \frac{\delta V - E}{R_a}$$

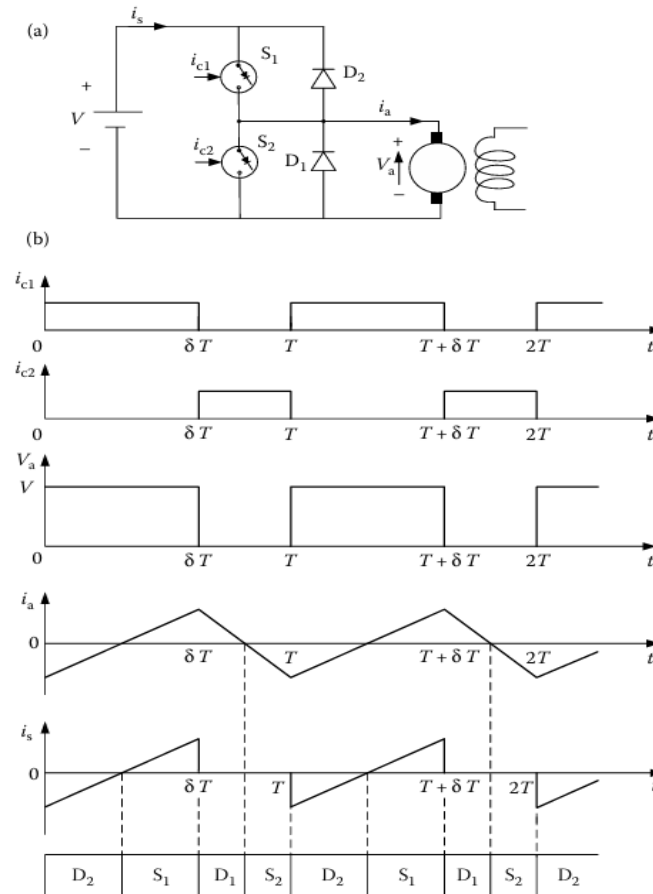


Figure 10: Forward motoring and regenerative braking control using class C two-quadrant chopper (a) Chopper circuits (b) waveforms

Above equation suggests that the motoring operation takes place when $\delta > (E/V)$, and that regenerative braking occurs when $\delta < (E/V)$. The no-load operation is obtained when $\delta = (E/V)$.

Four-Quadrant Operation

The four-quadrant operation can be obtained by combining two class C choppers as shown in Figure 6.11, which is referred to as a class E chopper. In this chopper, if S_2 is kept closed continuously, and S_1 and S_4 are controlled, a two-quadrant chopper is obtained, which provides positive terminal voltage (positive speed) and the armature current in either direction (positive or negative torque), giving a motor control in quadrants I and IV. Now if S_3 is kept closed continuously and S_1 and S_4 are controlled, one gets a two-quadrant chopper that can supply a variable negative terminal voltage (negative speed), and the armature current can be in either direction (positive or negative torque), giving a motor control in quadrants II and III.

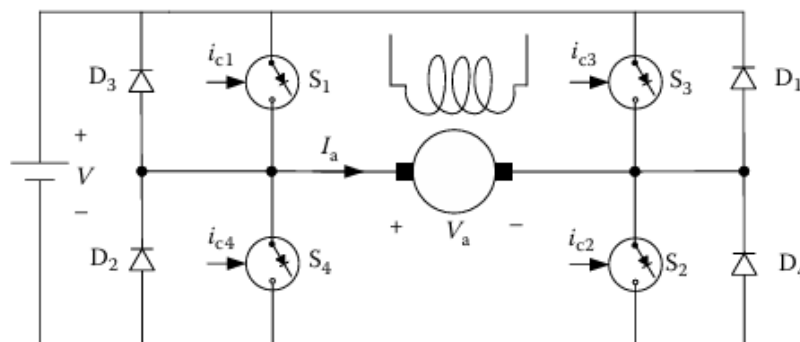


Figure 11: Class E four-quadrant chopper.

This control method has the following features:

The utilization factor of the switches is low due to the asymmetry in the circuit operation. Switches S_3 and S_2 should remain on for a long period. This may create commutation problems when the switches are using thyristors. The minimum output voltage depends directly on the minimum time for which the switch can be closed, since there is always a restriction on the minimum time for which the switch can be closed, particularly in thyristor choppers. The minimum available output voltage, and therefore the minimum available motor speed, is restricted. To ensure that switches S_1 and S_4 , or S_2 and S_3 are not on at the same time, some fixed time interval must elapse between the turn-off for one switch and the turn-on of another switch. This restricts the maximum permissible frequency of operation. It also requires two switching operations during a cycle of the output voltage.